MANAGING & INTERACTING WITH DOCKER CONTAINERS

DOCKER COMMANDS

1. docker ps ---------- to check the running containers
2. docker ps -a ---------- to check the running as well as existed containers
3. docker images ---------- to check the created images
4. docker image ls ---------- to check the images
5. docker search httpd ---------- to search the image in dockerhub
6. docker pull (httpd) ---------- to pull the image from dockerhub
7. docker run (httpd) ---------- the container run in foreground
8. docker run -d (httpd) ---------- the container run in background
9. docker kill (container ID) ---------- to kill the container
10. docker start (container ID) ---------- to start the container
11. docker stop (container ID) ---------- to stop the container
12. docker inspect (container ID)------- to check all the info about the container
13. docker run -d -p (httpd) ---------- to assign manual port to the container
14. docker run -d -P (httpd) ---------- to assign random port to container (32768-64000)
15. docker top (container ID) ---------- to get containers process related information
16. docker port (container ID) ----------to check the port of container
17. docker logs (container ID) ---------- to check the logs of a container
18. docker stop $(docker ps -q) ----------to stop all the running containers
19. docker rm $(docker ps -aq) ----------to remove all the existed container
20. docker rmi $(docker images -q) --------to remove all the images
21. docker rmi -f $(docker images -q) -----to remove images forcefully
22. docker rm -f (container ID) ---------- to remove the container
23. docker rename (1st name) (2nd name) ------to rename a container
24. docker stats (container ID) ----------to check the cpu utilization of every container

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DOCKER CONTAINER LIFECYCLE

1. docker start
2. docker stop
3. docker rm
4. docker prune

EXPOSE APPLICATION TO THE WORLD

1. docker run -d -p 8081:80 --name nginx_container --rm nginx
2. docker ps
3. docker inspect nginx_container | grep “IP”
   private_ip: 172.17.0.2
4. <public_ip:8081> -----> hit it on the web

Note:

1. -d ----------> run in detach mode
2. -p ----------> to assign a manual port
3. –name ----> to get the name to the container
4. –rm --------> automatically remove container after stop
5. Inspect ----> to get all the info about the container

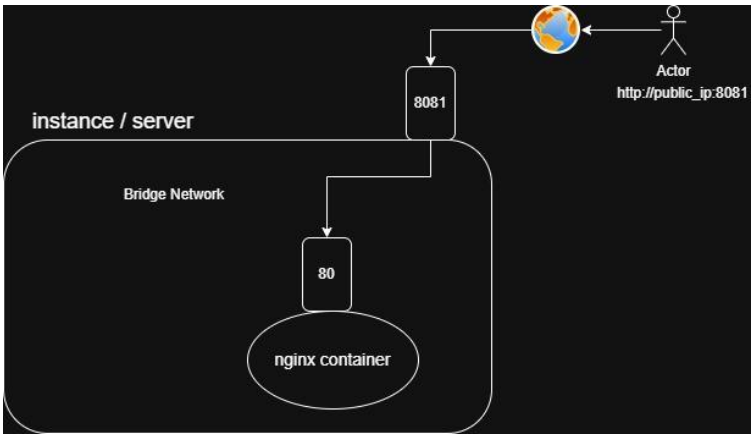
Port Explanation:

1. 8081 -----> server host port ----> port on your system
2. 80 --------> cont. port ----> port inside container

When someone hits port 8081 on my system, send that request to port 80 inside the container

How it works:

- You open browser → http://public_ip:8081
- Request goes to:
  - Host port 8081
  - Docker forwards it to container port 80
  - NGINX inside container responds
- Bridge Network is Docker’s default private network that allows containers to communicate internally and connect to the outside world through port mapping.



DOCKER EXEC

1.docker run -d -p 8081:80 –name nginx –rm nginx
2.docker ps

3.docker exec -it nginx /bin/bash

1. cd usr/share/nginx/html/
2. echo “hello nginx” > index.html
3. exit

7.docker ps
8.<public_ip:8081>

Note:

1. exec ----------> to go inside a running container
2. /bin/bash -----> Open Bash shell inside container
3. -it ----------> to get interactive terminal
4. -i ----------> keep input open
5. t ----------> to get the terminal

HOW TO CHANGE THE DEFAULT NGINX PAGE

1. docker run -d -p 8081:80 –name nginx –rm nginx
2. vim index.html
   Welcome to docker
3. docker cp index.html nginx:/usr/share/nginx/html/index.html
4. <public_IP:80811>
